Neha Rao

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SUMMARY

Computational biologist (M.S. Bioinformatics, Boston University, 2025) who builds and maintains **analysis pipelines and data tools on Linux HPC** for large omics datasets, with prior work in **GWU's HIVE Lab**, at NIH, and in DoD research. I write **Python, R, and shell** with **Snakemake and Nextflow**, have trained **PyTorch** models on cryo-electron tomography volumes, and validate ML and LLM outputs against experimental evidence. I care about the parts that keep a shared computing environment trustworthy: testing, QC, reproducibility, and documentation other people can follow.

EXPERIENCE

Bioinformatics Analyst | Baylor College of Medicine *Sep 2026 – Present*

- Analyze **single-cell, spatial, and multi-omics data** for the lab's own research and for collaborating groups across BCM, and contribute analyses to grant development.
- Build **computational and AI-driven tools** that let researchers without a coding background run and explore analyses.

Research Intern, HIVE Lab | The George Washington University *Jun 2026*

- Validated ML-extracted protein modification records against published evidence for GlyGen's GlycoSiteMiner.
- **Identified and escalated systematic data integrity issues** (accession mapping, residue notation) with reproducible record-level examples; contributed evaluation sets and feature engineering to pipeline v2.0.

STEAM Instructor | iCode School *Oct 2025 – Sep 2026*

- Developed curriculum and taught programming (Python, Lua, Scratch), databases, and robotics to students from elementary through high school, keeping daily hands-on practice in Python and SQL.
- Explained technical concepts to non-specialist audiences and debugged code live, the same skills used in documenting pipelines and supporting wet-lab collaborators.

Data Analyst, AEOP Fellow | Walter Reed Army Institute of Research *Feb 2025 – Sep 2025*

- *Term fellowship that ended early due to a lapse in federal funding.* Developed and maintained **modular analysis pipelines** across transcriptomic, metabolomic, lipidomic, and microbiome datasets on Linux HPC, from raw output through alignment, QC, normalization, and differential expression.
- Ran QC across concurrent projects, catching batch inconsistencies and normalization errors before they propagated downstream; summarized results for the team.

Cancer Research Intern, CRI-SP Fellow | National Cancer Institute, NIH *May 2024 – Aug 2024*

- **Downloaded, harmonized, and batch-corrected** TCGA and ICGC sequencing data while preserving biological signal.
- Built an **XGBoost model** to identify transcriptional phenocopies of the $KRAS^{G12D}$ mutation, extended it to single-cell RNA-seq, and analyzed spatial transcriptomics data; delivered as a one-command reproducible pipeline (NIH Poster Day 2024).

Database Development Intern, Hou Lab | Boston University *Jan 2024 – May 2024*

- Designed **database architecture and pipelines for loading, querying, and transforming multi-omics data** (bulk and single-cell ATAC-seq, RNA-seq, ChIP-seq) across cell types and disease states using Seurat, ArchR, and Scanpy.
- Built a framework for **cell-type-specific TF–CRE–gene regulatory networks** integrating epigenomic and transcriptional profiles, plus interactive locus-view tracks, GWAS network overlays, and R Shiny dashboards; supported two publications.

TECHNICAL SKILLS

- **Programming & HPC:** Python (Pandas, NumPy, scikit-learn, PyTorch), R/Bioconductor, Bash/shell scripting, SQL; Linux, Slurm, Conda; Git/GitHub
- **Workflows & Software:** Snakemake, Nextflow; pipeline testing, debugging, and documentation; Streamlit and R Shiny apps; Claude Code, GitHub Copilot
- **Bioinformatics:** sequence alignment (STAR, HISAT2, Bowtie2, BWA), SAMtools, BEDTools, MACS2; DESeq2, edgeR, limma; clustering and dimensionality reduction; Seurat, Scanpy, ArchR
- **Data Types:** bulk and single-cell RNA-seq, ATAC-seq, ChIP-seq, spatial transcriptomics, metabolomics, lipidomics, microbiome
- **Structural & ML:** cryo-ET deep learning (PyTorch, RELION); protein post-translational modification sites; XGBoost, cross-validation; LLM output validation
- **Data Quality:** QC, batch correction, public data harmonization (TCGA, ICGC, GEO), provenance tracking, FAIR principles

PUBLICATIONS & CERTIFICATIONS

- Co-author (**Rao, N.**) with Fu, T. T., Hou, L., et al. on two papers on epigenomic regulatory networks in brain disease, *European Neuropsychopharmacology* 87:8 (2024) and 99:234 (2025).
- CMMI Certified Practitioner (process improvement and quality management framework).

EDUCATION

M.S., Bioinformatics | Boston University, Boston, MA *2025*

B.Sc. with Honors, Biotechnology | M.S. Ramaiah University of Applied Sciences, Bangalore, India *2022*